HEART DISEASE PREDICTION USING MACHINE LEARNING

ADVANCED MACHINE LEARNING (DSC540)

FINAL PROJECT REPORT
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Submitted by

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ABSTRACT :
Heart disease is one of the leading causes of death worldwide, and detecting it early is key to improving patient outcomes. However, traditional diagnostic methods rely heavily on doctors' expertise, which can be slow and subjective, sometimes leading to delayed treatment. This project uses machine learning techniques to predict the risk of heart disease based on patient health data. By using the Cleveland Heart Disease dataset, the study focuses on data cleaning, exploration, feature selection, and applying machine learning models like Random Forest and Gradient Boosting, Logistic Regression, SVM, etc The goal is to provide doctors with accurate tools for early detection, ultimately improving patient care and reducing healthcare costs.

INTRODUCTION :
Heart disease is a serious global health problem, causing 32% of deaths worldwide, According to the WHO. In the United States alone, it caused 702,880 deaths in 2022, making up one in every five deaths. Coronary Artery Disease (CAD), the most common type of heart disease, affects 1 in 20 adults and is a major cause of heart attacks. In the US a heart attack happens every 40 seconds, adding up to 805,000 cases each year. Along with its high death rates, heart disease also creates a heavy financial burden on healthcare systems and families. These concerning statistics show the urgent need for better and faster ways to detect and treat heart disease.

Current methods for diagnosing heart disease mostly depend on doctors' expertise, which can take time and sometimes be influenced by personal judgement. These traditional methods often cause delays in identifying and treating patients at high risk, especially when symptoms are unclear or hard to notice. These delays can lead to missed chances for early treatment, increasing the risk of serious complications like heart attacks and heart failure. To address this, there is a growing need for data-driven solutions to help healthcare professionals make faster and more accurate diagnoses. This project uses machine learning (ML) techniques to predict heart disease risk by analysing patient health data like blood pressure, cholesterol levels, and exercise factors.

LITERATURE REVIEW :
There is a lot of research done in heart disease using machine learning. I have done literature reviews on different methods.

Elyan et al. utilised the Cleveland dataset, applying various preprocessing, different feature engineering techniques, and feature selection methods such as TF-IDF, correlation coefficients, and n-grams. Five different machine learning models were employed for training, including Naive Bayes, Support Vector Machines (SVM), Artificial Neural Networks, and a novel Hybrid Machine Learning (HML) approach. Among these, HML achieved the best predictive performance, with an accuracy of 99% on the Cleveland dataset [1].

Nashif et al, A cloud-based heart disease prediction system was designed, using various machine learning algorithms to enhance accuracy. Through a comparative analysis of ML models using the WEKA platform, Support Vector Machines (SVM) demonstrated high performance, achieving an accuracy of 97.53%, with sensitivity and specificity rates of 97.50% and 94.94%, respectively. The validation process incorporated 10-fold cross-validation on widely-used open-access datasets, ensuring robust evaluation of the model's performance [2].

Detrano et al, A new model was tested to estimate the likelihood of coronary artery disease in three different patient groups. This model, developed using data from 303 patients at the Cleveland Clinic, was applied to groups in Hungary, the United States, and Switzerland, showing different levels of disease and prediction accuracy. Compared to the CADENZA algorithm, the new model overestimated less in the Hungarian and American groups and slightly underestimated in the Swiss group, while CADENZA consistently overestimated. [3].

Sharma et al, In this study the authors used a UCI heart disease dataset with 14 different features related to heart disease and applied it on 4 different machine learning algorithms which include Random Forest, SVM, Naive Bayes, and Decision Tree. Random Forest gave higher accuracy with less runtime. But the authors, did not mention the evaluation method using cross validation and AUC [4] .

Bhatt et al, Various Machine learning models like Random Forest, Decision Tree, Multilayer Perceptron, and XGBoost were trained on a Kaggle dataset with 70,000 entries, using an 80:20 train-test split. After hyperparameter tuning with GridSearchCV, the Multilayer Perceptron model achieved the highest accuracy of 87.28% with cross-validation, outperforming the other models. All models also demonstrated strong performance with AUC values of 0.94–0.95, highlighting their effectiveness.

After reviewing the research on heart disease, I applied the methods mentioned in the literature to my study. This included using ensemble algorithms like Random Forest, GridSearchCV to find the best model fit, cross-validation, and feature engineering and selection techniques. These approaches helped me identify the most accurate model for predicting heart disease effectively.

Methodology :

Dataset Overview:

Data Source : https://archive.ics.uci.edu/dataset/45/heart+disease

The Cleveland Heart Disease dataset, obtained from the UCI Machine Learning Repository, contains 303 patient records with 14 medical features. These features include age, gender, blood pressure, cholesterol levels, exercise capacity, and other health metrics. The target variable, **"num"**, indicates heart disease presence and severity, where 0 denotes no disease, and values 1 indicate the Presence of the Heart disease.

Notable features include chest pain type, fasting blood sugar, electrocardiographic results, maximum heart rate, exercise-induced angina, and thalassemia. These metrics are crucial for analysing and predicting heart disease.

Data Preprocessing:

The data was cleaned and prepared for analysis by handling missing values, removing duplicates, and in the dataset the categorical variables were already converted into numbers, Class imbalance in the Data set.

Handling Missing Values : The dataset had some missing values in the "ca" and "thal" columns. The "ca" column was missing 4 values, and the "thal" column was missing 2 values. To fix this, the rows with missing data were removed, leaving no missing values in the dataset.

Checking Duplicates : The dataset was checked for any duplicate rows, but none were found. Therefore, no rows needed to be removed.

Handling Categorical and Numerical Features: The dataset comprises both categorical and numerical features, with categorical variables already encoded as numbers. Thus, no additional transformations were required for analysis.

Class Imbalance Dataset: The dataset includes 139 instances of heart disease and 164 instances of non-heart disease, with a ratio of 0.45 to 0.54. This indicates that the dataset is approximately balanced.

Binary Classification: The dataset originally contains 5 different classes, representing varying degrees of heart disease (1, 2, 3, 4) and no heart disease (0). I converted this into a binary classification, where any value from 1 to 4 is labelled as 1 (indicating heart disease), and 0 remains as 0 (indicating no heart disease).

Exploratory Data Analysis (EDA):

EDA is a Process to explore and Understand the Data to identify the Underlying Patterns in the Data, Identify potential problems like outliers and remove the outliers, It is Used to Understand the relationship among features and allows us to improve the Quality of the Data, It helps to find the relationship between predictor and target features, EDA also helps with feature selection and feature engineering like having correlation with the target variable, we consider those as important features, In EDA we performed correlation analysis and generated Heatmaps, box plots, detected outliers and also we Created plots for Data Distributions.

Data Distributions:

The below figure 1 represents the distributions of various features in the dataset, helps to understand the patterns and trends in the data. Features like age, trestbps (resting blood pressure), and thalach (maximum heart rate), followed approximately Normal distribution, with values concentrated around their means. However, some features, such as chol(cholesterol) and oldpeak (ST depression), showed skew distributions, indicating that most values are concentrated on one side while a few outliers exist on the other. Binary features like sex, fbs, (fasting blood sugar), and exang(exercise-induced angina) highlight that one category is significantly more frequent than the other, reflecting clear majority trends. Categorical variables like "cp" (chest pain type), "restecg" (resting ECG results), and "slope"

show that certain categories are more common, while others are less represented.. These visualisations provide a clear picture of the dataset's structure, highlighting any skewness, category imbalances, or dominant trends, which can guide further preprocessing and model-building steps.

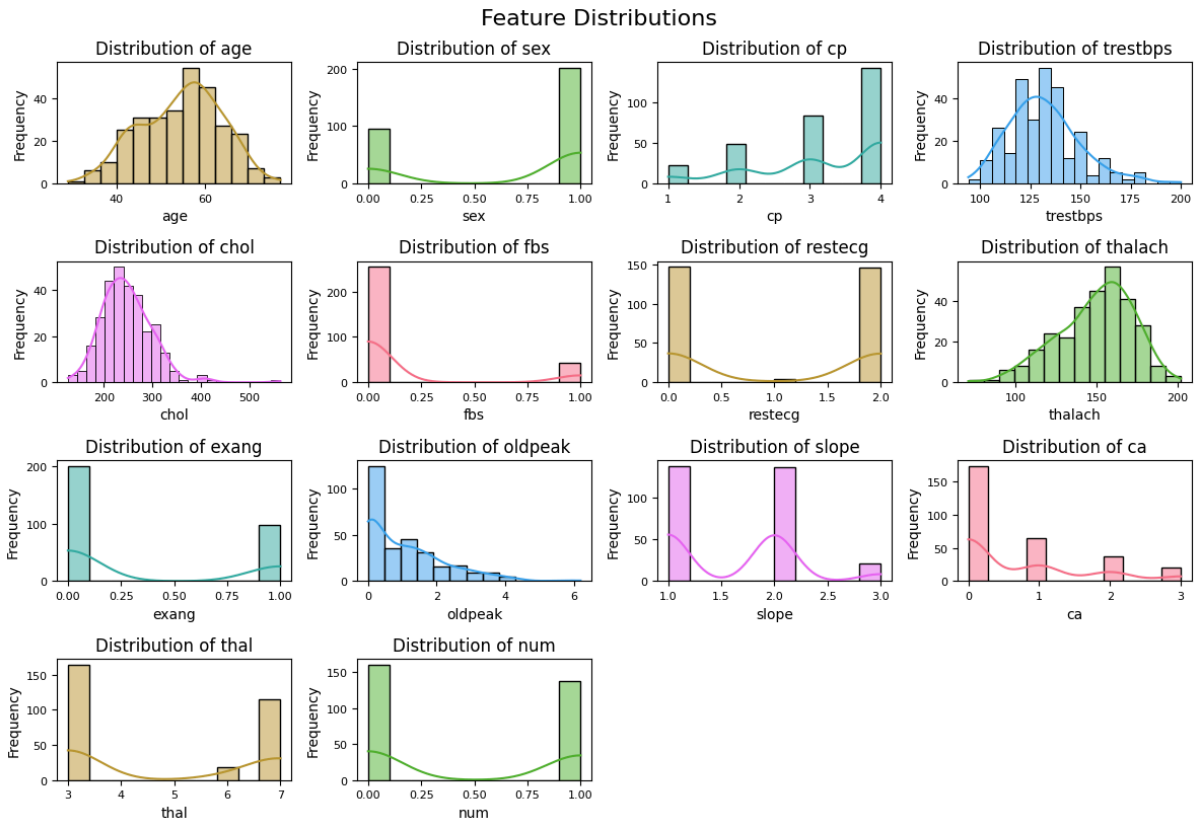

Fig1 : Data Distributions

Outliers:

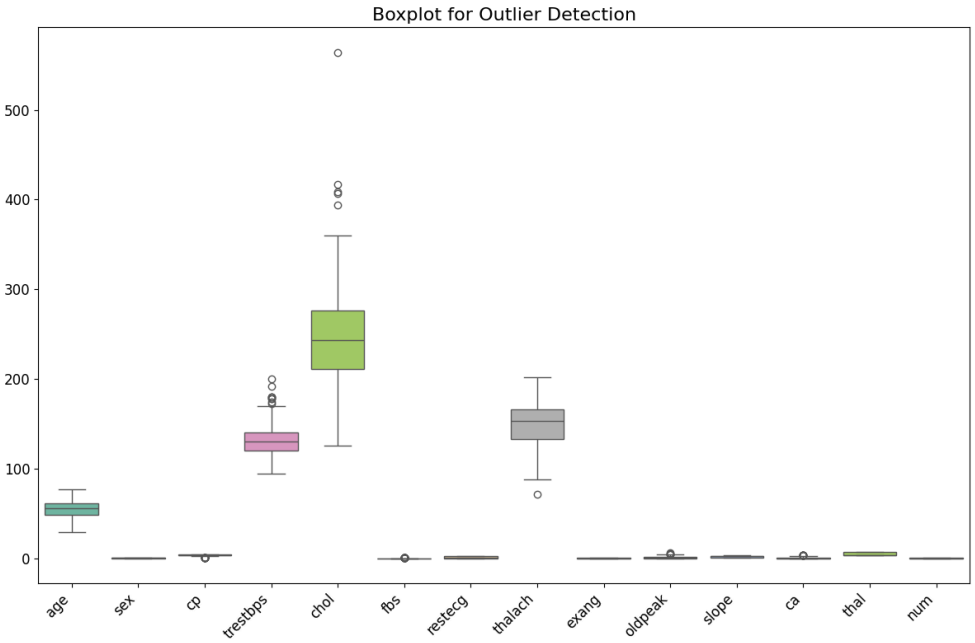

Fig2: Boxplot for Outlier Detection

An Outlier is a Datapoint that is completely different from the rest of the data, it lies far away from the majority of the data and outliers can be caused by various factors like measurement errors or data entry mistakes or unusual cases. Outliers can distort the statistics, mislead the data, they can negatively affect the model performance and lead to incorrect conclusions or false patterns in data analysis. Identifying and removing outliers is crucial for ensuring more accurate analysis. In the above Fig 2 box plot, outliers were identified as points outside the "whiskers". To detect these outliers, we used the Interquartile Range (IQR) method, which calculates the difference between the 75th and 25th percentiles (Q3 and Q1). We then defined the upper and lower bounds for outliers by adding or subtracting 1.5 times the IQR from Q3 and Q1. Data points outside these bounds were considered outliers and removed from the dataset. The IQR method is reliable because it is less sensitive to extreme values compared to standard deviation based methods. The cleaned dataset is now free of outliers and ready for further analysis, enhancing the accuracy and reliability of the data for analysis or machine learning models.

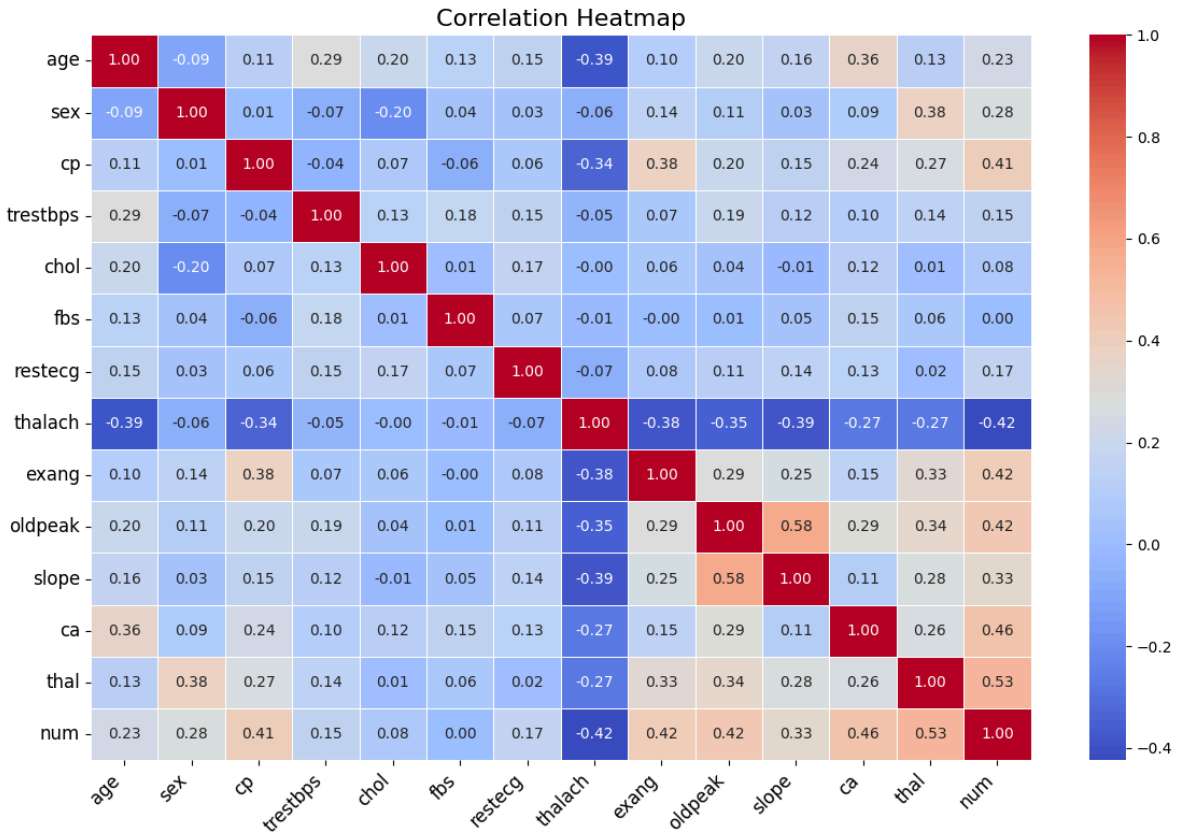

Fig3 : Correlation Heatmap.

Correlation analysis helps find relationships between variables and shows how strongly they are connected. It can help predict one variable from another and uncover patterns that might not be obvious. This is useful in machine learning for selecting important features. However, correlation does not mean one variable causes the other, and it mainly focuses on linear relationships, so it might not capture nonlinear ones well.

In the above Fig3 Correlation Heatmap, The highest correlation is between slope and oldpeak (0.579), which shows a correlation, but not strong enough to consider them redundant and remove one. The target variable num is moderately correlated with ca (0.463), thal (0.527),

and oldpeak (0.424), which could help improve model performance. Additionally, num and thalach (-0.424) have an inverse relationship, where higher thalach values are linked to lower num values. These features ca, oldpeak, and thalach could positively impact the model.

Feature Engineering :
Before using the dataset for training machine learning algorithms, it is important to standardise all the features to the same range using StandardScaler. This is because the features in the dataset have different value ranges. If we train a machine learning model directly on such data, the model might prioritise features with larger values and give less importance to those with smaller values. This issue is especially common in algorithms that rely on distance calculations. To avoid these problems and ensure fair treatment of all features, we standardise the dataset. After standardising the data the model training will also take less time. Therefore, standardisation was done using StandardScaler from scikit-learn library.

Feature Selection:
The feature selection was performed using Recursive Feature Elimination (RFE) in combination with a Random Forest Classifier. RFE is a technique that selects the most important features by recursively eliminating less important ones. The importance of features is determined using the Random Forest model, which assigns importance based on how effectively each feature contributes to improving the classification performance.

The Random Forest Classifier was configured with 200 decision trees (n_estimators=200) and used the entropy criterion to measure split quality. The RFE algorithm was set to retain a specific number of features (n_features_to_select) by progressively removing features with the least importance in steps of 10% (step=0.1) until the desired number of features was reached.

Selected Features: ['age', 'sex', 'cp', 'restecg', 'thalach', 'exang', 'oldpeak', 'slope', 'ca', 'thal']
These features were selected because they provide the most valuable information for predicting the target variable (num) and have the greatest impact on the model's performance. By reducing the dataset to these features, the model's complexity is minimised while retaining its predictive accuracy.

Model Training :
Model training is the process of helping a machine learning algorithm learn patterns from data so it can make accurate predictions. The data is split into two parts: the training set, which the algorithm uses to learn, and the testing set, which is used to check how well the model works on new, unseen data. During this process, the algorithm adjusts itself to reduce mistakes and improve its predictions. The goal is to build a model that generalises and works well not only on the data it has been trained on but also on new data, making it reliable and useful in real-world situations.

We have used a Range of Machine Learning Algorithms, each one offering a unique approach to make Predictions. We have Used Random Forest which is an ensemble method that combines several Decision trees to enhance the accuracy and reduce the overfitting by averaging their results, Similarly, Gradient Boosting builds decision trees in sequence, with each new tree aimed at addressing the mistakes of the previous ones, using gradient descent to minimising errors.Support Vector Machine (SVM) is a strong algorithm that works well in complex situations by finding the best way to separate different classes of data. K-Nearest Neighbors (KNN) is simpler and classifies data based on the majority class of its closest neighbours, without needing to build a model. Logistic Regression is used for binary classification, estimating the chances of data belonging to a specific class. AdaBoost is a technique that combines multiple weak models, like simple decision trees, and focuses on the points that were misclassified to improve overall accuracy. Together, all these algorithms provided various methods to solve the problem and helped in choosing the best one.

Cross Validation Hyperparameter tuning :

Cross-validation is a method used to test how well a model can perform on new, unseen data. It works by splitting the data into several smaller parts, or "folds." The model is trained on the (k - 1) folds and tested on the remaining kth fold. This process is repeated for each fold, so the model gets tested on all different unseen subsets of the data. Cross-validation helps to get a more accurate estimate of a model's ability to generalise and reduces the risk of overfitting, which happens when the model only works well on the training data but not on new data.

Grid Search CV is a technique for finding the best hyperparameters for a model. This helps in the process of hyperparameter tuning in an easy way. Grid Search tests all possible combinations of these settings and uses cross-validation to see which combination performs the best. Hyperparameter tuning, through methods like Grid Search, improves the model by adjusting these settings to ensure it not only works well on the training data but also performs effectively on new, unseen data, making it more reliable for predictions.

Results :

Model	**Accuracy**	**AUC**	**CV Runtime**
Random Forest	0.8667	0.9520	0.1673
Gradient Boosting	0.8000	0.9107	0.1366
SVM	0.8667	0.9598	0.0119
KNN	0.8500	0.9503	0.0019
Logistic Regression	0.8500	0.9487	0.0057
Ada Boost	0.8167	0.9319	0.1951

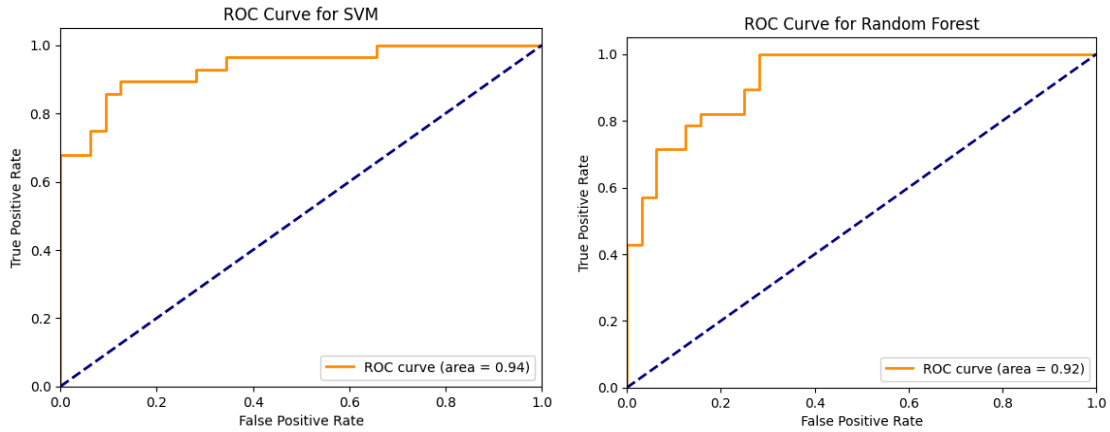

Fig4: ROC Curve

Among all the models, SVM (Support Vector Machine) is the best because it combines high accuracy and AUC. It has an accuracy of 0.8667, which is the same as Random Forest, making it one of the top models. Its AUC score of 0.9598 is the highest, showing that SVM is the best at correctly distinguishing between different classes, even if the data is uneven. Additionally, SVM has the fastest training time of 0.0119 seconds, which makes it efficient.These results suggest that SVM is great at making accurate predictions on new, unseen data. The high AUC shows it works well across various thresholds, and the accuracy ensures it performs reliably. With a good balance of strong performance and efficiency, SVM is the best choice for this task and is expected to work well on new data.

Discussions: Even though the dataset is small with only 303 rows, models like SVM and Random Forest performed well with the existing features, achieving around 0.95 AUC and 0.86 accuracy, with negligible runtime. The methodology and techniques from past research and the literature review significantly contributed to improving the model's performance. I applied Recursive Feature Selection to identify the top 10 features that had the most impact on boosting the model. The selected features are ['age', 'sex', 'cp', 'restecg', 'thalach', 'exang', 'oldpeak', 'slope', 'ca', 'thal'], which were crucial for the model. Since the dataset is relatively balanced, AUC and accuracy were calculated, along with Precision, Recall, and F1 scores, which were also high but not included in the paper.

Random Forest and SVM may have worked well, the reason could be due to their ability to handle complex and high-dimensional data effectively. Random Forest is an ensemble learning method that builds multiple decision trees and combines their results, which helps in reducing overfitting and improving generalisation, especially with a small dataset like this. SVM, on the other hand, is effective in high-dimensional spaces and works well for classification tasks by finding the optimal hyperplane that separates different classes. Both models are robust to noise and can manage both linear and non-linear relationships in the data, making them well-suited for predicting heart disease based on the available features.

KNN may not have worked well due to its sensitivity to irrelevant features and the small size of the dataset, which can lead to overfitting or poor generalisation. Logistic regression might

have underperformed because it assumes a linear relationship between features and the target, which may not capture the complex patterns in the heart disease data and also most of the data are categorical variables.

Conclusions: Heart disease is a leading cause of mortality worldwide, often caused by factors such as high blood pressure, high cholesterol, and a lack of physical activity. Early prediction and diagnosis of heart disease are crucial for effective treatment and prevention. Finally, the machine learning approach for the heart disease prediction study achieved strong performance, with both the SVM and Random Forest models reaching 86% accuracy after thorough analysis and the selection of 10 key features. These results demonstrate the effectiveness of the chosen models and feature selection process in predicting heart disease. This study highlights the potential of machine learning models in improving heart disease diagnosis. With further improvements, these models could become valuable tools in real-world healthcare settings.

Future work: Future work for this study could focus on improving predictive performance by increasing the dataset size, which currently contains only 303 instances. Adding a few thousand more instances for the features that have the most impact on predicting heart disease could boost the model's performance, making it more suitable for real-world scenarios. The model could then be updated with real-time data and retrained periodically. Additionally, exploring advanced deep learning techniques could further improve the model's accuracy and performance. The model could also be deployed in real-world settings with a pipeline that allows it to update and retrain at regular intervals, ensuring it stays up to date.

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